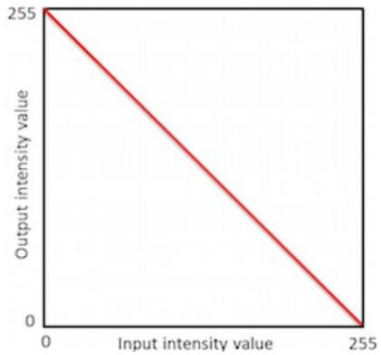
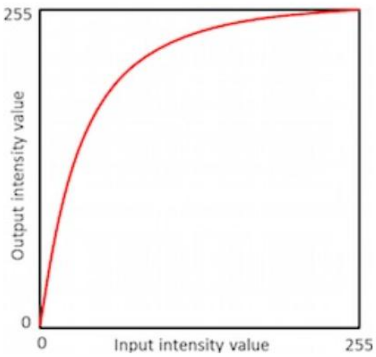
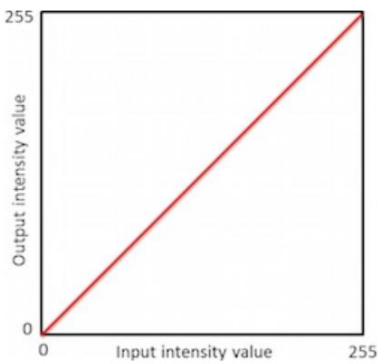
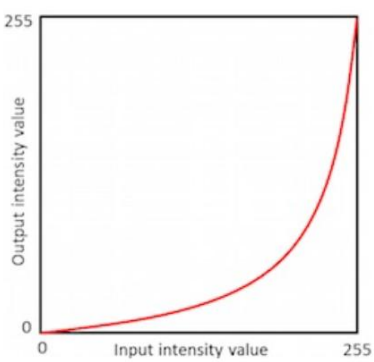
Homework #2

1. If you wanted to make an image look brighter than what it currently does, which one of the following intensity transformations would you use? **(5pts)**









2. The mean and standard deviation of pixel intensity values in an 8-bit gray-scale image are 120 and 10, respectively. What are the mean and standard deviation of pixel intensity values in the negative of this image? **(5pts)**

- 135, 10
- 120, 10
- 110, 20
- They can't be determined without knowing the size of the image.

3. Check all statements that are true regarding image histogram equalization **(5pts)**:

- [ ] After histogram equalization, the intensity values are more effectively distributed over the histogram range.
- [ ] The mean intensity value of the pixels in an image increases after histogram equalization.
- [ ] After histogram equalization images will always look brighter.
- [ ] If pixel p is the brightest pixel in image x, it will remain the brightest pixel after histogram equalization.

4. Consider a 64x64 image with 7 gray levels 0, 1, 2, 3, 4, 5, 6. The histogram of this image is given by:

k	r_k	n_k
0	0	148
1	1/6	300
2	2/6	500
3	3/6	2,200
4	4/6	500
5	5/6	300
6	1	148

Find a point transformation $z = H(r)$ that when applied on image f produces an image g with histogram that is close to the following:

k	r_k	n_k
0	0	498
1	1/6	550
2	2/6	600
3	3/6	800
4	4/6	600
5	5/6	550
6	1	498

Calculate the new histogram so obtained. **(20pts)**

5. Convolve the image I with the kernel M as shown below assuming free boundary condition, i.e. for a pixel near the edge, if any part of the kernel extends beyond the image boundary, assume the image values outside are 0 (zeros). **(15pts)**

$$I = \begin{bmatrix} 8 & 1 & 2 & 1 & 7 \\ 9 & 3 & 10 & 4 & 0 \\ 1 & 5 & 10 & 9 & 8 \\ 9 & 10 & 5 & 8 & 9 \\ 6 & 10 & 8 & 10 & 7 \end{bmatrix} \quad M = \begin{bmatrix} 0 & 0 & 1 & 0 & 0 \\ 0 & 0 & 1 & 0 & 0 \\ 1 & 1 & 1 & 1 & 1 \\ 0 & 0 & 1 & 0 & 0 \\ 0 & 0 & 1 & 0 & 0 \end{bmatrix}$$

6. If *F(u, v)* is the Fourier transform of the discrete image *f*(*m,n*), Prove that the Fourier transform of the discrete image *f*(*am* + *bn*, *cm* + *dn*) is *F*(*du-cv*, *av-bu*), where *ad* − *bc* = 1, and *a, b, c, and d* are integers. **(15 pts)**

7. For each signal length of $N = 2^3, 2^4, \dots, 2^{13}$, generate a random signal and compute its discrete Fourier transform (DFT) using two methods:

(1) direct matrix multiplication, $X = W_N x$, where $\left(W_N\right)_{k,n} = e^{-j2\pi kn/N}$.

(2) the Fast Fourier Transform (FFT).

Measure the execution time of both methods, plot runtime versus N, and plot the speedup $T_{\text{direct}}/T_{\text{FFT}}$ versus N. Compare the measured trends with the theoretical computational complexities, $O(N^2)$for direct DFT and $O(N\log\ N)$for FFT, and discuss how the advantage of FFT changes as the signal size increases. Additionally, generate and plot the real and imaginary components of the DFT matrix for N = 100. **(15pts)**

8. Image Compression via the 2D Fast Fourier Transform: To understand the sparsity of images in the frequency domain and explore how the 2D Discrete Fourier Transform (DFT) can be utilized for lossy image compression. You will implement an algorithm to compress an image by retaining only a fraction of its most significant Fourier coefficients and quantitatively evaluate the resulting image quality. (20 pts)

(1) The Fast Fourier Transform (FFT) reveals the frequency components of an image. In many natural images, most of the energy is concentrated in a small number of coefficients. Load your chosen image and convert it to a double-precision grayscale matrix. Compute the 2D FFT of the image. Sort the Fourier coefficients by their absolute magnitude in descending order. Create a compression routine that retains only the few percentage of largest magnitude coefficients, setting all other coefficients to zero. Apply the inverse 2D FFT to reconstruct the image. **Deliverable:** Generate a 2x2 subplot showing the original image alongside the reconstructed images retaining 5%, 1%, and 0.2% of the largest coefficients.

(2) Visual inspection is subjective. We will use Mean Absolute Error (MAE) to measure how many intensity levels a pixel is off by, on average. Loop your compression algorithm across retention percentages 0.1%, 0.5%, 1%, 2%, 5%, 10%, 20%, 50%, 100%.Calculate the MAE between the original and reconstructed images for each retention percentages. **Deliverable:** Plot MAE on the y-axis retention percentages on the x-axis (use a logarithmic scale for the x-axis). Looking at your plot, describe how the error behaves as retention drops below 5%. Is the degradation linear or exponential?

(3) Instead of picking the largest coefficients anywhere in the spectrum, a low-pass filter only keeps frequencies near the center (the DC term). Implement a circular ideal low-pass filter. Calculate the exact number of coefficients retained in your 1% magnitude thresholding (from Part 1), and set the radius of your low-pass filter so it retains that exact same number of pixels. Reconstruct the image using this low-pass filter. **Deliverable**: A side-by-side plot of the 1% Magnitude Threshold reconstruction and the 1% Low-Pass Filter reconstruction.Which method produces a sharper image? Explain mathematically why this happens based on what high-frequency components represent in an image.